

1. Design Calculations

1.1 Compensation Capacitor Selection

The load capacitance is:

$$C_L = 10 \text{ pF}$$

For a two-stage CMOS op-amp, the compensation capacitor is chosen as approximately 20% of the load capacitance.

$$C_c \approx 0.2 \times C_L$$

$$C_c \approx 0.2 \times 10$$

$$C_c = 2 \text{ pF}$$

Therefore,

$$\mathbf{C_c = 2 \text{ pF}}$$

1.2 Tail Current Calculation

The required slew rate is:

$$SR = 10 \text{ V}/\mu\text{s}$$

For a Miller compensated two-stage op-amp,

$$SR = I_5 / C_c$$

Therefore,

$$I_5 = SR \times C_c$$

$$I_5 = (10 \times 10^6)(2 \times 10^{-12})$$

$$I_5 = 20 \times 10^{-6} \text{ A}$$

$$I_5 = 20 \text{ }\mu\text{A}$$

Therefore,

$$\mathbf{I_5 = 20 \text{ }\mu\text{A}}$$

1.3 Differential Pair Transconductance

The gain-bandwidth relationship is:

$$g_{m1} = 2\pi(\text{GBW})C_c$$

Substituting:

$$\text{GBW} = 5 \text{ MHz}$$

$$C_c = 2 \text{ pF}$$

$$g_{m1} = 2\pi(5 \times 10^6)(2 \times 10^{-12})$$

$$g_{m1} = 62.83 \times 10^{-6} \text{ S}$$

$$g_{m1} = 62.83 \text{ } \mu\text{S}$$

Therefore,

$$\mathbf{g_{m1} = 62.83 \text{ } \mu\text{S}}$$

1.4 Sizing of M1 and M2

The tail current divides equally between M1 and M2.

$$I_{D1} = I_{D2} = I_5 / 2$$

$$I_{D1} = I_{D2} = 20 / 2$$

$$I_{D1} = I_{D2} = 10 \text{ } \mu\text{A}$$

Using:

$$g_{m2} = 2K_n' I_{D2} (W/L)$$

Rearranging:

$$(W/L) = g_{m2}^2 / (2K_n' I_{D2})$$

Substituting:

$$(W/L) = (62.83^2) / [2 \times 120 \times 10]$$

$$(W/L) = 3947.6 / 2400$$

$$(W/L) = 1.644$$

Therefore,

$$(W/L)_1 = (W/L)_2 = 1.644$$

1.5 Sizing of M3 and M4

From the DC operating point:

$$V_{SG3} = 2.5 - 1.804$$

$$V_{SG3} = 0.696 \text{ V}$$

The overdrive voltage is:

$$V_{OV3} = V_{SG3} - |V_{TP}|$$

$$V_{OV3} = 0.696 - 0.5$$

$$V_{OV3} = 0.196 \text{ V}$$

Using:

$$I_D = \frac{1}{2}K_p'(W/L)V_{OV}^2$$

Substituting:

$$10 = 0.5 \times 25 \times (W/L) \times (0.196)^2$$

$$10 = 0.48(W/L)$$

$$(W/L) = 20.8$$

A practical value was selected:

$$(W/L)_3 = (W/L)_4 = 20$$

1.6 Sizing of M5

From simulation:

$$V_{GS5} = 0.774 \text{ V}$$

The overdrive voltage is:

$$V_{OV5} = V_{GS5} - V_{TN}$$

$$V_{OV5} = 0.774 - 0.5$$

$$V_{OV5} = 0.274 \text{ V}$$

Using:

$$I_D = \frac{1}{2}K_n'(W/L)V_{OV}^2$$

Substituting:

$$20 = 0.5 \times 120 \times (W/L) \times (0.274)^2$$

$$20 = 4.50(W/L)$$

$$(W/L) = 4.44$$

Therefore,

$$(W/L)_5 = 4.44$$

1.7 Sizing of M8

To ensure proper current mirroring:

$$(W/L)_8 = (W/L)_5$$

Therefore,

$$(W/L)_8 = 4.44$$

1.8 Bias Resistor Calculation

From simulation:

$$V_{GS8} = 0.774 \text{ V}$$

Therefore,

$$V_R = V_{DD} - V_{GS8}$$

$$V_R = 2.5 - 0.774$$

$$V_R = 1.726 \text{ V}$$

For a reference current of:

$$I_{REF} = 20 \text{ } \mu\text{A}$$

$$R = V_R / I_{REF}$$

$$R = 1.726 / (20 \times 10^{-6})$$

$$R = 86.3 \text{ k}\Omega$$

A practical resistor value was selected:

$$\mathbf{R = 86 \text{ k}\Omega}$$

1.9 Initial Design of M6

To obtain a phase margin close to 60° , the second-stage transconductance is chosen as:

$$g_{m6} = 10g_{m1}$$

$$g_{m6} = 10 \times 62.83$$

$$g_{m6} = 628.3 \text{ } \mu\text{S}$$

Using:

$$g_m^2 = 2K_p'ID(W/L)$$

Assuming:

$$I_{D6} = 80 \text{ } \mu\text{A}$$

$$(W/L)_6 = (628.3^2) / [2 \times 25 \times 80]$$

$$(W/L)_6 = 98.7$$

Therefore,

$$\text{Initial } (W/L)_6 = 98.7$$

1.10 Initial Design of M7

Using first-order design equations:

$$\text{Initial } (W/L)_7 = 20.77$$

1.11 Simulation Optimization

The initial values obtained from analytical calculations were:

- $(W/L)_6 = 98.7$
- $(W/L)_7 = 20.77$

AC simulations showed that the gain requirement was not fully satisfied. Therefore, iterative optimization was performed in LTspice.

Theoretical sizing equations provide only first-order estimates and do not account for channel-length modulation, parasitic capacitances, finite output resistance, and frequency-dependent effects. Therefore, M6 and M7 were optimized through simulation to achieve the required gain and phase margin.

Final values:

Transistor	Initial W/L	Final W/L
M6	98.7	234.5
M7	20.77	24.8

The optimized design achieved:

- DC Gain = 68.47 dB
- Unity Gain Bandwidth = 4.87 MHz
- Phase Margin = 73.24°
- Bias Current $\approx 20 \mu\text{A}$

Final Transistor Sizing

Transistor	W/L
M1	1.644
M2	1.644
M3	20
M4	20
M5	4.44
M6	234.5
M7	24.8
M8	4.44